CISPA
HELMHOLTZ CENTER FOR
INFORMATION SECURITY

HPC Cluster Guide

TRUSTWORTHY MACHINE LEARNING 2026

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ACCESS

- Each team will be assigned **one shared cluster account** (username and password).
- You can find this under your CMS Personal Status tab.
- Use this account for all your cluster work throughout the course.
- The cluster is only reachable from within the university network or via VPN.

VPN Setup: https://serviceportal.hiz-saarland.de/esc?id=emp_taxonomy_topic&topic_id=e31ae340c3da6a5098fa9dcf0501311e

ACCESS - Setup Video

- You can find the video recording here -

https://drive.google.com/file/d/1BbIU_Y_p17ExrJ_A1tv0v9KPiIEWSPhB/view?usp=sharing

ACCESS

- If you are already a team of 2, your cluster login details are already available on you CMS Personal Status page under the heading **cluster_credentials**

atml_teamXXXXX → this is your username
Rest is the password.

ACCESS

Two ways to connect:

- SSH (terminal):
  a. *ssh <username>@conduit.hpc.uni-saarland.de* **or**
  b. *ssh <username>@conduit2.hpc.uni-saarland.de*

- VS Code Remote (More accessible):
  a. SSH Connect to *conduit.hpc.uni-saarland.de* or *conduit2.hpc.uni-saarland.de* via the Remote SSH extension

Either way, you will be required to enter the password provided to you.

ACCESS

Save this to ~/.ssh/config for easier access -

No need to type your username every time you SSH or connect via VS Code Remote SSH. Add the following to your SSH config file:

Host conduit2.hpc.uni-saarland.de
HostName conduit2.hpc.uni-saarland.de
User <your-team-username>

After saving, you can connect simply with:
ssh conduit2.hpc.uni-saarland.de

```
Host conduit2.hpc.uni-saarland.de
    HostName conduit2.hpc.uni-saarland.de
    User atml_team099
```

Getting the Assignment Files

Do NOT use git clone on the cluster. The .pt files use Git LFS which is not available on the cluster.

Instead, download directly:

```
mkdir ~/tml26_task1
cd ~/tml26_task1

wget "https://huggingface.co/datasets/SprintML/tml26_task1/resolve/main/pub.pt"
wget "https://huggingface.co/datasets/SprintML/tml26_task1/resolve/main/priv.pt"
wget "https://huggingface.co/datasets/SprintML/tml26_task1/resolve/main/model.pt"
wget "https://huggingface.co/datasets/SprintML/tml26_task1/resolve/main/task_template.py"
```

Running Jobs - Submit File

Save this as mia.sub (replace <username> with your team username):

```
universe = docker
docker_image = pytorch/pytorch:2.3.1-cuda12.1-cudnn8-devel
executable = /opt/conda/bin/python
arguments = /path-to/task_template.py
output = runlogs/mia.$(ClusterId).$(ProcId).out
error = runlogs/mia.$(ClusterId).$(ProcId).err
log = runlogs/mia.$(ClusterId).log
should_transfer_files = YES
request_GPUs = 1
request_CPUs = 1
request_memory = 4G
requirements = UidDomain == "cs.uni-saarland.de"
+WantGPUHomeMounted = true
+WantScratchMounted = true
queue 1
```

Running Jobs - Commands

Submit a job:
mkdir -p runlogs → (only once)
condor_submit mia.sub

Check your jobs: *condor_q*

Cancel a job: *condor_rm <job_id>*

Cancel all your jobs: *condor_rm -a*

Job states:

- Idle → waiting for a free worker node
- Running → executing on a worker node
- Held → something is wrong, check with: *condor_q -hold <id>*

Interactive Jobs (for debugging)

- Test your code interactively before submitting a full job.
- Gets you a live shell inside the Docker container with GPU access.

condor_submit -i mia.sub

Once inside:
cd /home/<username>/tml26_task1
python task_template.py

Note: Interactive jobs are killed automatically after 1 hour. Use it only for debugging. For long runs, always submit a normal job.

Important Notes

- Use absolute paths in your script.
- Relative paths break inside the container.

Use this pattern instead:

```
from pathlib
import Path
BASE = Path(__file__).parent
PUB_PATH = BASE / "pub.pt"
PRIV_PATH = BASE / "priv.pt"
MODEL_PATH = BASE / "model.pt"
```